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Myocardial and Renal Renin–Angiotensin System Enzyme Activity in Dogs

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ABSTRACT

Background: Methods to quantify the activity of the tissue renin–angiotensin system (RAS) are needed.

Objective: To estimate myocardial and renal RAS enzyme activity in postmortem tissues.

Animals: Convenience sample of 15 purpose-bred laboratory dogs with experimental heartworm infection (Group 1) euthanized as part of an unrelated study and 22 client-owned dogs which died or were euthanized for any reason (Group 2).

Methods: Prospective study where myocardial and renal tissue samples were frozen within 20 min of euthanasia (Group 1) or during routine necropsy exam (Group 2). Homogenized tissues were incubated with spiked angiotensin I or II (AngII) under control and inhibitor conditions to assess relative enzyme contributions to AngII or angiotensin 1–7 formation. Freezing was delayed (2- or 7-h in Groups 1 and 2, respectively) in paired samples, to evaluate the effect of time to freezing on RAS enzyme activity.

Results: Samples were harvested within 20 min of euthanasia (Group 1) or a median of 21.9 h after death or euthanasia (Group 2). RAS enzymes were active in all samples. The median contribution of angiotensin-converting enzyme to AngII formation was greater than 88% in all tissues. The 90% confidence limits of the geometric mean of the ratio of the angiotensin production of paired samples in 10/14 (Group 1) and 16/19 (Group 2) experiments met equivalence requirements (lower bound > 0.8 and upper bound < 1.2).

Conclusions and Clinical Importance: Renin–angiotensin system enzymes are present and active in canine myocardial and renal tissues for greater than 24 h postmortem.

1 | Introduction

The renin–angiotensin system (RAS) regulates blood pressure and electrolyte homeostasis and is upregulated in response to hypovolemia, hypotension, inadequate sodium intake, and sustained sympathetic nervous system activation. Circulating RAS activity is increased in dogs with congestive heart failure (CHF) [1–4]; blockade of the RAS with angiotensin-converting enzyme inhibitors (ACEi) alone and in combination with aldosterone antagonists (spironolactone) improves survival [5–12]. Conversely, in the long

Abbreviations: ACE, angiotensin-converting enzyme; ACE2, angiotensin-converting enzyme 2; ACEi, angiotensin-converting enzyme inhibitor; Ang, angiotensin; AP, angiotensin peptide; CV, coefficient of variation; IQR, interquartile range; IVS, interventricular septum; LC–MS, liquid chromatography-mass spectrometry; LLOQ, lower limit of quantification; LVFW, left ventricular free wall; MMVD, myxomatous mitral valve disease; MR, mitral regurgitation; NEP, neutral endopeptidase; PCP, prolylcarboxypeptidase; PEP, prolylendopeptidase; RAAS, renin–angiotensin–aldosterone system; RAS, renin–angiotensin system; RVFW, right ventricular free wall; UCD-VMTH, University of California, Davis Veterinary Medical Teaching Hospital; ZPP, Z-pro-prolinal.

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preclinical periods of diseases such as myxomatous mitral valve disease (MMVD), administration of ACEi with and without spironolactone has not been shown to delay the time to CHF or to prolong survival [13–15]. In dogs with chronic kidney disease (CKD), circulating RAS activity is higher when compared to life-stage- and sex-matched controls [16]. In cats with CKD, renal angiotensinogen gene transcript levels and angiotensin I (AngI) concentrations are higher when compared to control cats [17]. Determining the ideal approach to RAS modulation in these chronic diseases requires an improved understanding of circulating and tissue RAS.

Angiotensin peptide (AP) concentrations are low and often below the lower limit of quantification (LLOQ) in canine myocardium extirpated within 60 min of death (Oyama unpublished data). Whether this is due to rapid AP metabolism or inherently low AP levels in dogs is unknown. To explore this question, this study utilizes postmortem samples collected as rapidly as possible (within 20-min of euthanasia) from purpose-bred laboratory dogs involved in an unrelated research project that were infected with the canine heartworm (*Dirofilaria immitis*). In contrast to AP concentrations, angiotensin-converting enzyme (ACE) activity remains stable in postmortem sheep brain [18] and human myocardial [19] tissues for hours. This study takes advantage of postmortem enzyme stability and uses a novel approach to estimate tissue RAS enzyme activity by measuring the formation rates of angiotensin II (AngII) and angiotensin 1–7 (Ang1-7) in tissues incubated with AP precursors. The formation rates of AngII and Ang1-7 reflect the relative balance between key enzymes of the classical (ACE, chymase) and alternative (angiotensin-converting enzyme 2; ACE2) RAS. Additional enzymes that counter-regulate the classical RAS by breaking down AngI and AngII include prolyl-carboxypeptidase (PCP), prolyl-endopeptidase (PEP), and neprilysin (NEP; Figure 1). In addition to reducing AngII levels, some of the metabolites (e.g., Ang1-7) resulting from the activity of these enzymes also counteract the pathologic effects of AngII [20]. Increased tissue AngII formation relative to Ang1-7 formation suggests a dominance of the classical RAS, a situation that has been found in humans with CKD [21].

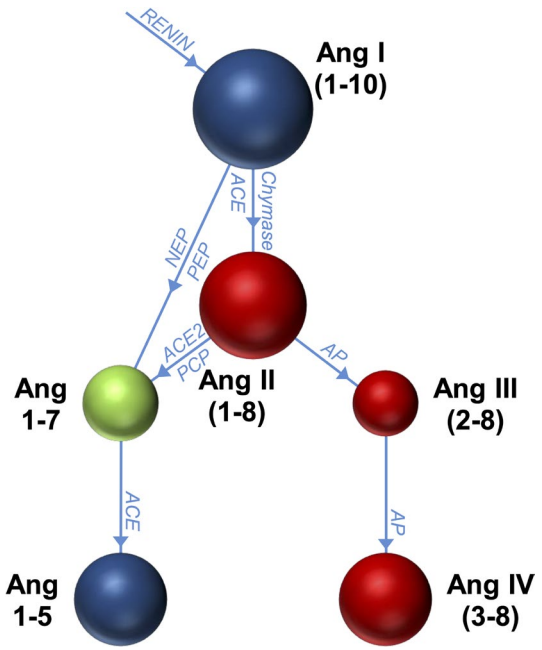

FIGURE 1 | Angiotensin peptide metabolism, including angiotensin (Ang) peptides AngI, AngII, Ang1-7, and Ang1-9 and their catalysts. ACE, angiotensin-converting enzyme; ACE2, angiotensin-converting enzyme 2; NEP, neprilysin; PCP, prolyl-carboxypeptidase; PEP, prolyl-endopeptidase.

The two primary objectives of this study were to estimate RAS enzyme activity in postmortem myocardial and renal tissues and to evaluate enzyme stability by repeating enzyme activity assessment on samples where freezing was delayed. A third objective was to determine if APs are quantifiable in quickly (10–20 min postmortem) extirpated tissue samples. We hypothesized that RAS enzyme activities could be estimated in postmortem myocardial and renal tissue, and that enzyme activity would be stable within the first few hours of death. We hypothesized that most APs would be above the assay LLOQ in quickly harvested samples.

2 | Materials and Methods

2.1 | Dogs—Group 1

Fifteen purpose-bred, specific-pathogen-free dogs infected with *Dirofilaria immitis* were euthanized as part of an unrelated study where they served as untreated controls [22]. All dogs had eight male and 12 female adult heartworms introduced via transjugular transplantation. Dogs were randomly divided into three groups of five dogs each and euthanized 75-, 105-, and 135-day post-worm transplantation.

2.2 | Dogs—Group 2

Twenty-two client-owned dogs that died or were euthanized at the UC Davis Veterinary Medical Teaching Hospital (UCD-VMTH) for any reason and deceased dogs whose bodies were donated by community members to the UCD-VMTH were recruited. Owner permission was given through signed authorization for an unrestricted necropsy. Dogs whose bodies were refrigerated within 8 h of death and who had tissue samples collected within 40 h of death were included. There were no restrictions on previous medical history, medication, breed, body weight, sex, neuter status, or age.

2.3 | Myocardial Tissue Collection—Groups 1 and 2

Left ventricular free wall (LVFW) samples (500 mg in Group 1 and 250 mg in Group 2) were collected adjacent to the midpoint of the paraconal coronary artery and extending apically. Care was taken not to include visible blood vessels. A second 250 g sample was taken from the mid-right ventricular free wall (RVFW; Group 1) and mid-interventricular septum (IVS; Group 2). In both groups, a single wedge (500 mg in Group 1 and 250 mg in Group 2) from the midpoint of the convex border, including cortex and medulla, was collected from the kidney.

In Group 1, tissues were immediately collected after euthanasia, rinsed with phosphate-buffered saline, and frozen in liquid nitrogen. A second left ventricular and kidney sample remained

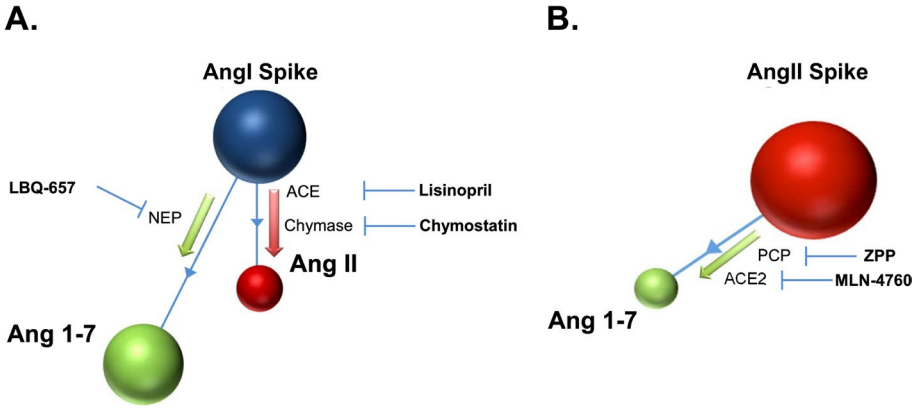

FIGURE 2 | A) Production rates of angiotensin II and 1–7 (AngII, Ang1-7) were evaluated in tissue homogenates spiked with (A) angiotensin I (AngI) or (B) angiotensin II (AngII) and subjected to sequential additions of enzyme inhibitors. See text for details. ACE, angiotensin-converting enzyme; ACE2, angiotensin-converting enzyme 2; NEP, neprilysin; PCP, prolyl-carboxypeptidase; ZPP, Z-pro-prolinal.

at room temperature for 2 h (+2-h sample) before transfer to a cassette and storage at −80°C. In Group 2, initial samples were collected at the time of necropsy. In this group, the time since death (natural or euthanasia) varied. The initial samples were rinsed with tap water, transferred to cassettes, and frozen at −80°C. In 10 dogs from Group 2, a second sample was placed in a refrigerator (4°C) for 7 h (+7-h sample) before being transferred to a −80°C freezer.

2.4 | Tissue Angiotensin Peptide Concentrations—Group 1

Angiotensin peptides were quantified by Attoquant Diagnostics (Vienna, Austria) as described previously [23]. Briefly, frozen myocardium was homogenized using a mortar and pestle under liquid nitrogen. The tissue powder was dissolved at 100 mg/mL in 6 mol/L aqueous guanidinium chloride and 1% (v/v) trifluoroacetic acid by cooled sonication using a 2 mm microtip (Sonics and Materials, Newton, New Jersey). Stable isotope-labeled internal standards for individual angiotensin (Ang) metabolites (AngI, AngII, Ang1-7, Ang1-5, AngIII, and AngIV) were added to tissue homogenates at 200 pg/mL and stored at −80°C until liquid chromatography with tandem mass spectrometry analysis (LC–MS). After C18-based solid-phase extraction, samples were subjected to LC–MS analysis using a reversed-phase analytical column (Acquity UPLC C18; Waters, Milford, Massachusetts) operating in line with a XEVO TQ-S triple quadrupole mass spectrometer (Waters) in MRM mode. Internal standards were used to correct for matrix effects and peptide recovery of the sample preparation procedure for each angiotensin metabolite in each individual sample. Angiotensin peptide concentrations were calculated considering the corresponding response factors determined in appropriate calibration curves, on condition that integrated signals exceeded a signal-to-noise ratio of ten.

2.5 | Tissue Angiotensin II and Angiotensin 1–7 Formation Rates—Groups 1 and 2

Endogenous enzyme activity was evaluated based on AngII and Ang1-7 formation rates using previously described methods [24]. Before the spike of AngI or AngII, tissue homogenates were dissolved in phosphate-buffered saline (pH 7.4) supplemented with $ZnCl_2$ (10 μmol/L). Samples were also subjected to several preexperiments (Supporting Information).

AngI spike: Dissolved tissue homogenates were spiked with AngI (10 μg/mL) and the inhibitors Amastatin (aminopeptidase inhibitor, 10 μmol/L), MLN-4760 (ACE2 inhibitor, 10 μmol/L), and ZPP (PCP/PEP inhibitor, 20 μmol/L) to prevent degradation of the spiked AngI and its products AngII or Ang1-7. After incubation at 37°C for 30 min in the presence and absence of selective inhibitors, AngII and Ang1-7 were measured via LC–MS/MS, and the formation rates (pg/μg protein/h) were calculated. In order to elucidate the contribution of the enzymes of interest (ACE, chymase, NEP) to product formation, selective inhibitors were added in four approaches: (1) no additive (control), (2) chymostatin (chymase inhibitor, 10 μmol/L), (3) chymostatin and lisinopril (ACE inhibitor, 10 μmol/L), or (4) chymostatin, lisinopril, and LBQ-657 (NEP inhibitor, 10 μmol/L; Figure 2A).

AngII spike: Dissolved tissue homogenates (see above) were spiked with AngII (10 μg/mL) and the inhibitors Amastatin (aminopeptidase inhibitor, 10 μmol/L) and lisinopril (ACE inhibitor, 10 μmol/L), to prevent degradation of the spiked AngII or its product Ang1-7. Spiked tissue homogenates were incubated under three conditions: (1) no additive (control), (2) ZPP (PCP/PEP inhibitor, 20 μmol/L), or (3) ZPP and MLN-4760 (ACE2 inhibitor, 10 μmol/L; Figure 2B). Ang1-7 formation rates were measured for all conditions. The assay coefficient of variation (CV) for angiotensin formation rate quantification from tissue homogenates within the quantifiable range was 10%.

2.6 | Ante- and Postmortem Cardiac and Renal Diagnoses—Group 2

Date and time of death, antemortem diagnoses, medical treatments, and the time between euthanasia and sample acquisition were recorded. Clinical, physical, and postmortem examination findings suggestive of cardiac or renal disease in the absence of an antemortem diagnosis were also recorded. These included the presence of a left-sided apical systolic heart murmur, the presence of a ≥1-week history of azotemia (creatinine ≥1.6) or

proteinuria (urine protein-to-creatinine ratio; UP:C ≥ 0.5), and the presence of cardiac or renal anatomic or histopathologic abnormalities. Degenerative changes to the mitral valve were assigned a grade of 0 through 4, based on the Whitney classification scheme [25].

2.7 | Statistical Analysis

Data were evaluated for normality using the Shapiro–Wilk test. Data are presented as median and interquartile range (IQR). The tissue concentrations of AngI and AngII were compared among the three tissue types (LVFW, RVFW, and kidney) in the Group 1 samples using the Kruskal–Wallis test. The AngII and Ang1-7 formation rates were compared between the LVFW and RVFW (Group 1) and LVFW and IVS (Group 2) using the Mann–Whitney test. The AngII and Ang1-7 formation rates were also compared between control (no enzyme inhibition) and different enzyme inhibition scenarios using the Mann–Whitney test or Kruskal–Wallis test (with Dunn's multiple-comparison posttest) depending on whether two or three experiments were performed, respectively. Simple linear regression was used to evaluate the relationship between directly measured AP concentrations and AP production rates in the control experiment in kidney samples from Group 1 dogs. The specific activity of each enzyme was calculated by determining each enzyme's percent contribution to AP formation. For example, the percent conversion of AngI to AngII by chymase was calculated as the chymostatin inhibition-sensitive fraction ([control minus inhibitor]/control × 100).

Ang1-7 to AngII production rates are presented as a unitless ratio and consist of the Ang1-7 formation rate (from AngII substrate) to AngII formation rate (from AngI substrate) during the control experiments. The Ang1-7 to AngII formation rate ratios were compared between sampling sites in both groups using the Mann–Whitney test.

The equivalence of AP formation rates and enzyme activity between paired samples (initial sample versus delayed sample) was compared using a paired t-test on the ratios (initial sample/delayed sample). The geometric mean and its 90% confidence limit were calculated, and equivalence was set at less than 20% variance (lower bound of 0.8 and upper bound of 1.2). There are no preestablished criteria for equivalence of this type of paired tissue sample, and a variance of 20% was slightly more stringent than what has been recommended by the Food and Drug Administration for bioequivalence studies [26]. The Fisher's exact test was used to evaluate the difference in equivalency rate between Groups 1 and 2. Values from the two timepoints in both groups were also compared using the Wilcoxon matched-pairs signed-rank test.

Angiotensin peptide concentrations that were below the LLOQ for the LC–MS assay were reported and graphed as the LLOQ/2. Statistical analysis was performed by commercial statistical software programs (GraphPad Prism 10 and SAS version 9.4). Statistical significance was set at ≤ 0.05.

3 | Results

3.1 | Dogs and Tissue Samples—Group 1

Samples were collected from 15 one-year-old purpose-bred specific pathogen-free dogs (seven male and eight female) with a median (IQR) weight of 8.8 kg (8.2–9.8). All dogs developed detectable *Dirofilaria immitis* microfilaremia and antigenemia before euthanasia at Days 75, 105, and 135 post-adult worm transplantation. No dog developed clinical signs related to heartworm infection during this time. The time between death and freezing tissues in liquid nitrogen was less than 20 min for all dogs. Paired samples were harvested and kept at room temperature for 2 h before freezing.

3.2 | Dogs and Tissue Samples—Group 2

Samples from 22 dogs of varying breeds were collected at the necropsy exam. Cases included 11 neutered males, six spayed females, two intact males, and three intact females with a median (IQR) age of 8.5 years (4.8–11.3) and median (IQR) weight of 19.9 kg (6.2–31; Table S1). The bodies of all dogs were refrigerated within 8 h of death. The median (IQR) time between death and freezing (−80°C) of tissue samples was 21.9 h (8.5–25.5). This sample served as the "time 0" sample for comparison to samples where freezing was delayed. Twenty-two samples of LVFW, 21 samples of IVS, and 21 samples of kidney were submitted for analysis. One dog did not have an IVS sample collected, and one dog did not have a kidney sample collected. Duplicate samples from the LVFW ($n = 10$), IVS ($n = 10$), and kidney ($n = 9$) were collected and remained in a refrigerator (4°C) for 7 additional hours before freezing (80°C).

3.3 | Tissue Angiotensin Peptide Concentrations—Group 1

Tissue AngI, AngII, and/or Ang1-7 concentrations were above the LLOQ of the LC–MS/MS assay in 6/15 LVFW samples. Angiotensin III, AngIV, and Ang1-5 were not detected in any of the LVFW samples. AngI, AngII, and/or AngIII were above the LLOQ in 7/15 RVFW samples. Ang1-7, Ang1-5, and AngIV were not detected in any RVFW samples. AngI and AngII concentrations were above the LLOQ in 15/15 kidney samples, Ang1-7 in 8/15 samples, AngIII in 11/15 samples, and AngIV in 1/15 samples; Ang1-5 was not detected in any kidney sample (Table 1). Concentrations of AngI and AngII were compared among the three tissue sites and were significantly higher in the kidney as compared to the two myocardial sites ($p = 0.001$ and $p = 0.02$ for AngI and AngII, respectively). Concentrations of AngI and AngII were compared among the three euthanasia groups (Days 75, 105, and 135 post-adult worm transplantation) and did not differ significantly.

3.4 | Tissue Production of AngII From AngI, Ang1-7 From AngII, and Ang1-7 From AngI—Groups 1 and 2

The addition of AngI to the tissue homogenates (LVFW, RVFW, and kidney for Group 1 and LVFW, IVS, and kidney for Group

TABLE 1 | Angiotensin (Ang) peptide concentrations (median, range) in homogenates of the left ventricular free wall (LVFW), right ventricular free wall (RVFW), and kidney from 15 dogs (Group 1).

	AngI (1–10) fmol/g	AngII (1–8) fmol/g	Ang1-7 fmol/g	AngIII (2–8) fmol/g	AngIV (3–8) fmol/g	Ang1-5 fmol/g
LVFW	33.7 (25.5–47.6); $n=5$	122.5 (33.8–168.4); $n=3$	32.9[a]	All < 10	All < 5	All < 10
RVFW	26.4 (26.1–61.5); $n=3$	67.6 (21.9–241.2); $n=4$	All < 10	85.6[a]	All < 5	All < 5
Kidney	619.2 (33.10–1847); $n=15$	226.7 (110.3–505.3); $n=14$	238.4 (104.2–727.7); $n=8$	123.8 (31.5–191.2); $n=11$	15.6[a]	All < 5

Note: Only values above the assay's lower limit of quantification are included. The number of measurable samples is indicated below the values. Samples were collected within 20 min of euthanasia.
[a]Only a single dog had a measurable sample.

TABLE 2 | Angiotensin (Ang) production rate under various conditions for left ventricular free wall (LVFW), right ventricular free wall (RVFW), and kidney less than 20 min after death for 15 dogs (Group 1); median (interquartile range).

Angiotensin production rate (pg/μg protein)/h	Control	Chymostatin	Chymostatin + lisinopril	Ang1-7:AngII[a]
AngII formation from AngI: LVFW	40 (33–53)	35 (30–48)	All < 5	24 (19–31)
AngII formation from AngI: RVFW	46 (28–58)	45 (24–56)	All < 5	19 (13–28)
AngII formation from AngI: kidney	656 (336–805)	632 (337–825)	All < 5	32 (29–56)[b]
Angiotensin Production Rate (pg/μg protein)/h	**Control**		**ZPP**	**ZPP + MLN-4760**
Ang1-7 formation from AngII: LVFW	1035 (980–1084)		343 (209–372)	23 (13–26)
Ang1-7 formation from AngII: RVFW	813 (714–914)		197 (147–236)	16 (10–18)
Ang1-7 formation from AngII: kidney	23 054 (18106–24 209)		19 937 (15122–21 498)	719 (549–839)
Angiotensin production rate (pg/μg protein)/h	**Control**		**Chymostatin + lisinopril + LBQ-657**	
Ang1-7 formation from AngI: LVFW	1990 (1841–2141)		1813 (1641–1934)	
Ang1-7 formation from AngI: RVFW	2208 (2043–2286)		2121 (1987–2132)	
Ang1-7 formation from AngI: kidney	3690 (3324–3944)		2441 (2133–2583)	

Note: Please see Figures 2, 3, and 4 for significant differences between enzyme inhibition experiments.
Abbreviation: ZPP, Z-pro-prolinal.
[a]The Ang1-7-to-AngII ratio was determined from formation rates of these two peptides during control incubations with AngI (for AngII production) and AngII (for Ang1-7 production) using the immediately frozen sample.
[b]The Ang1-7-to-AngII ratio differed significantly from the myocardial samples.

2) led to the formation of AngII (Tables 2 and 3 and Figure 3). When the chymase inhibitor, chymostatin, was added to tissue homogenates spiked with AngI, there was no significant change in AngII formation rates. When additionally the ACEi, lisinopril, was added to tissue homogenates spiked with AngI, there was inhibition of AngI to AngII production in all tissues (all values were below the LLOQ of 5 [pg/μg of tissue]/h).

The addition of AngII to the tissue homogenates led to the formation of Ang1-7 (Tables 2 and 3 and Figure 4). When the PCP inhibitor ZPP was added to tissue homogenates spiked with AngII, Ang1-7 formation was significantly reduced in the myocardial tissue homogenates but not in the kidney tissue homogenates. When additionally, the ACE2 inhibitor MLN-4760 was added to tissue homogenates spiked with AngII, there was inhibition of metabolism of AngII to Ang1-7 in most myocardial samples (values were below the LLOQ for nearly all dogs). In the kidney, the addition of ZPP and MLN-4760 led to a marked reduction in Ang1-7 formation as compared to control. Angiotensin peptide formation did not differ between the two myocardial sites.

The addition of AngI to the tissue homogenates also led to the formation of Ang1-7 (Tables 2 and 3 and Figure 5). In the myocardial samples, when the neprilysin inhibitor LBQ-657, chymostatin, and lisinopril were added to the tissue homogenates spiked with AngI, there was only a slight decrease in the Ang1-7 formation rate, though this was significant in the Group 1 samples. In the kidney, there was a significant reduction in the Ang1-7 formation rate in Groups 1 and 2.

TABLE 3 | Angiotensin (Ang) production rate under various conditions for left ventricular free wall (LVFW), interventricular septum (IVS), and kidney collected a median of 21.9 h after death for 22 dogs (Group 2); median (interquartile range). One dog did not have an IVS sample and one dog did not have a kidney sample collected.

Angiotensin production rate (pg/μg protein)/h	Control	Chymostatin	Chymostatin + lisinopril	Ang1-7:AngII[a]
AngII from AngI: LVFW	32 (25–48)	30 (24–41)	All dogs < 5	20 (12–39)
AngII from AngI: IVS	35 (28–43)	32 (23–38)	All dogs < 5	19 (12–35)
AngII from AngI: kidney	366 (251–717)	344 (246–720)	All dogs < 5	52 (35–79)[b]
Angiotensin production rate (pg/μg protein)/h	**Control**	**ZPP**	**ZPP + MLN-4760**	
Ang1-7 from AngII: LVFW	625 (411–986)	292 (174–469)	All but 3 dogs < 5 (12.0, 18.9, 24.7)	
Ang1-7 from AngII: IVS	745 (406–1006)	332 (187–499)	All but 3 dogs < 5 (14.1, 15.4, 22.0)	
Ang1-7 from AngII: kidney	21 908 (16199–27 382)	19 606 (13017–24 561)	824 (526–1011)	
Angiotensin production rate (pg/μg protein)/h	**Control**	**Chymostatin + lisinopril + LBQ-657**		
Ang1-7 from AngI: LVFW	2001 (1800–2207)	1963 (1741–2122)		
Ang1-7 from AngI: IVS	2015 (1860–2260)	1954 (1769–2155)		
Ang1-7 from AngI: kidney	3443 (3094–4279)	2224 (1694–2705)		

Note: Please see Figures 2, 3, and 4 for significant differences between enzyme inhibition experiments.
Abbreviation: ZPP, Z-pro-prolinal.
[a]The Ang1-7-to-AngII ratio was determined from formation rates of these two peptides during control incubations with AngI (for AngII production) and AngII (for Ang1-7 production).
[b]The Ang1-7-to-AngII ratio differed significantly from the myocardial samples.

3.5 | Enzyme Contributions to AngII and Ang1-7 Metabolism—Groups 1 and 2

In both groups where AngI was supplied as substrate, ACE accounted for the majority of AngII formation in all tissues, with chymase playing a smaller role (Tables 4 and 5). In both groups where AngII was supplied as substrate, PCP accounted for the majority of Ang1-7 formation in myocardial samples, with ACE2 playing a smaller role. In the kidney, ACE2 was responsible for most Ang1-7 formation in both Group 1 and Group 2. As Ang1-7 production occurred despite the inhibition of PCP and ACE2, other enzymes catalyzed Ang1-7 formation from AngII in the myocardial and renal tissues. In both groups where AngI was supplied as substrate, neprilysin contributed to a small amount of Ang1-7 formation in the myocardial samples and a moderate amount of Ang1-7 formation in the kidney. As Ang1-7 production occurred despite inhibition of chymase, ACE, and neprilysin, other enzymes catalyzed Ang1-7 formation from AngI in the myocardial and renal tissues.

3.6 | Myocardial Versus Renal Angiotensin Production Rates—Groups 1 and 2

In the control experiments, AngII production from AngI in the kidney was approximately 15 and 11 times (Groups 1 and 2, respectively) greater than AngII production from AngI in the myocardium. AngII formation rates did not differ between myocardial sites in either group (LVFW vs. RVFW in Group 1 and LVFW vs. IVS in Group 2).

In the control experiments, Ang1-7 production from AngII in the kidney was approximately 25 and 32 times (Groups 1 and 2, respectively) greater than Ang1-7 production from AngII in the myocardium. In Group 1, Ang1-7 formation rates were lower in the RVFW compared to the LVFW in the control and ZPP inhibitor experiments ($p < 0.001$ and $p = 0.003$, respectively). The Ang1-7 formation rates did not differ between the LVFW and IVS in Group 2.

In the control experiments, Ang1-7 production from AngI was nearly two times greater in the kidney as compared to the myocardial tissues. In Group 1, the Ang1-7 formation rate was higher in the RVFW as compared to the LVFW in the three-inhibitor (chymostatin, lisinopril, and LBQ-657) experiment ($p = 0.03$). The Ang1-7 from AngI formation rate did not differ between the LVFW and IVS in Group 2.

3.7 | Ratio of Ang1-7-to-AngII Formation Rates and Correlation Between AngII Concentration and Production Rate

The ratio of Ang1-7-to-AngII formation rates demonstrated that Ang1-7 was produced at a higher rate than AngII in all tissue types (Tables 2 and 3). The directly measured AngII concentrations from the kidney samples of Group I dogs were moderately correlated to the AngII production rate from AngI in these samples ($r^2 = 0.59$, $p = 0.0008$). AngII concentrations at the myocardial sites and Ang1-7 concentrations at all sites were frequently below the LLOQ, precluding analysis.

3.8 | Change in Angiotensin Formation Rates Over Time—Groups 1 and 2

In Group 1, AngII and Ang1-7 formation rates did not differ significantly between the samples frozen within 20 min and the

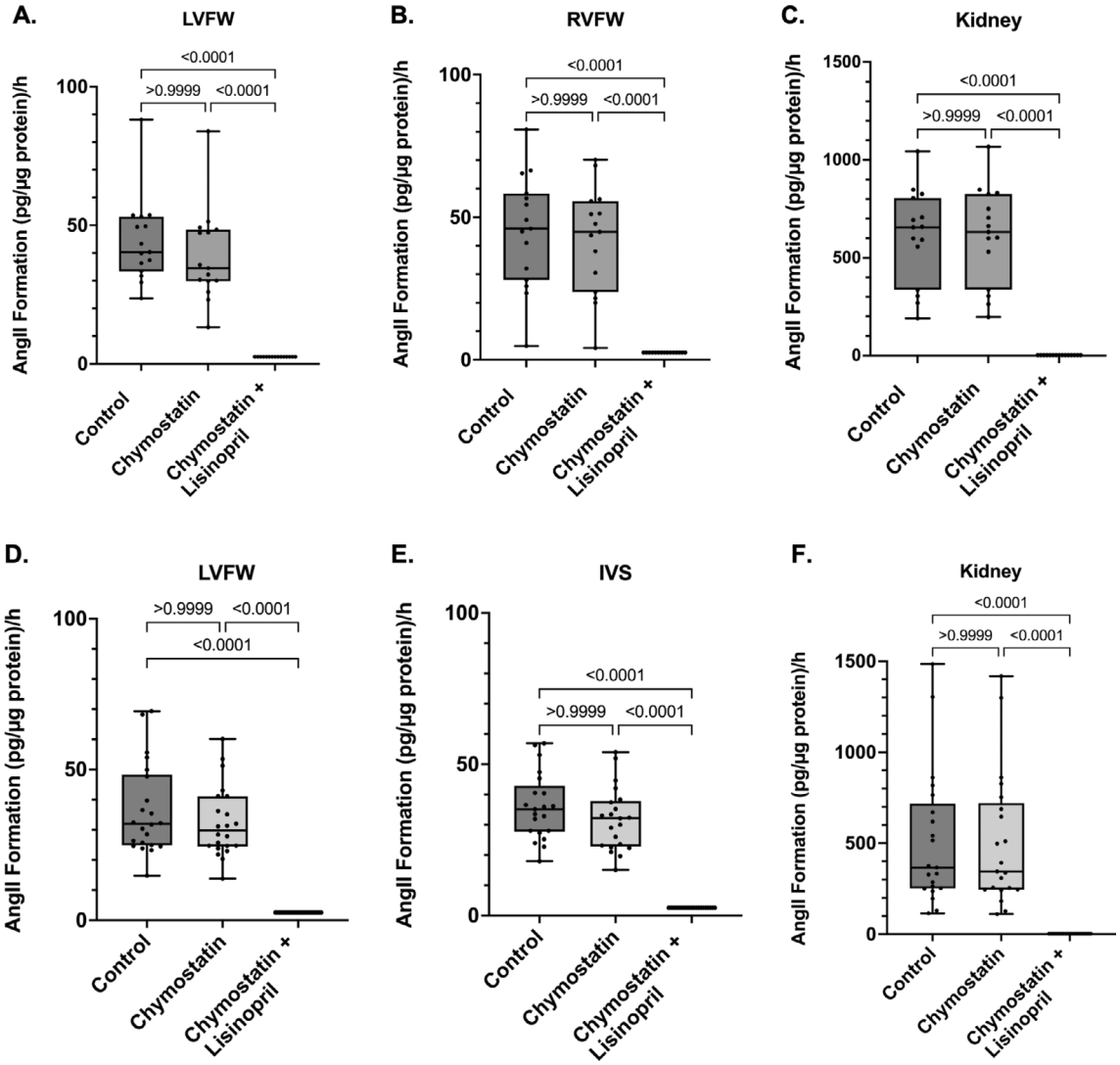

FIGURE 3 | Angiotensin II from angiotensin I formation rates ([pg/µg protein]/hour) in the myocardial and renal samples from dogs in Group 1 (A, B, and C) and Group 2 (D, E, and F). Homogenized tissues were incubated with angiotensin I alone (control) and with the chymase inhibitor chymostatin or chymostatin plus the angiotensin-converting enzyme inhibitor lisinopril. AngII, angiotensin II; IVS, interventricular septum; LVFW, left ventricular free wall; RVFW, right ventricular free wall. *Note: p* values are adjusted for multiplicity; *p* values less than 0.01 are approximate.

samples where freezing was delayed for 2 h (Table 6). In Group 2, the LVFW AngII production rates for the control and chymostatin experiments were significantly lower in the initial frozen samples as compared to the samples where freezing was delayed for 7 h. There were no other significant differences in AngII or Ang1-7 formation rates in any other experiment in Group 2 (Table 7).

In Group 1, the 90% confidence limits of the geometric mean of the ratio of the 0-h and +2-h samples in 10/14 experiments met the requirement of a lower bound > 0.8 and an upper bound < 1.2, and the samples were considered equivalent (Table 8). Two of the nonequivalent results were from myocardial tissue samples, and two were from kidney tissue samples. In Group 2, the 90% confidence limits of the geometric mean of the ratio of the 0-h and +7-h samples in 16/19 experiments met the requirement of a lower bound > 0.8 and an upper bound < 1.2, and the samples were considered equivalent (Table 9). Two of the nonequivalent results were from myocardial tissue samples, and one was from a kidney tissue sample. The rate of equivalence did not differ between the two groups ($p = 0.42$).

In Group 1, the 90% confidence limits of the geometric mean of the ratio of the percent enzyme contribution of the 0-h and +2-h samples in eight of the 13 experiments met the criteria for equivalence (Table 4 and Table S2). In Group 2, the 90% confidence limits of the geometric mean of the ratio of the percent enzyme contribution of the 0-h and +7-h samples in 13 of the 19 experiments met the criteria for equivalence (Table 5 and Table S3).

3.9 | Ante- and Postmortem Cardiac and Renal Diagnoses—Group 2

Four dogs had an antemortem diagnosis of cardiac disease and six dogs had an antemortem diagnosis of kidney disease. Cardiac diseases included MMVD (American College of Veterinary Internal Medicine Stage [4] B1 in one dog; Stage C in one dog), heart base mass (chemodectoma; one dog), and complete atrioventricular block (one dog). Death in three dogs was related to their underlying cardiac disease (Table S1). Kidney diseases

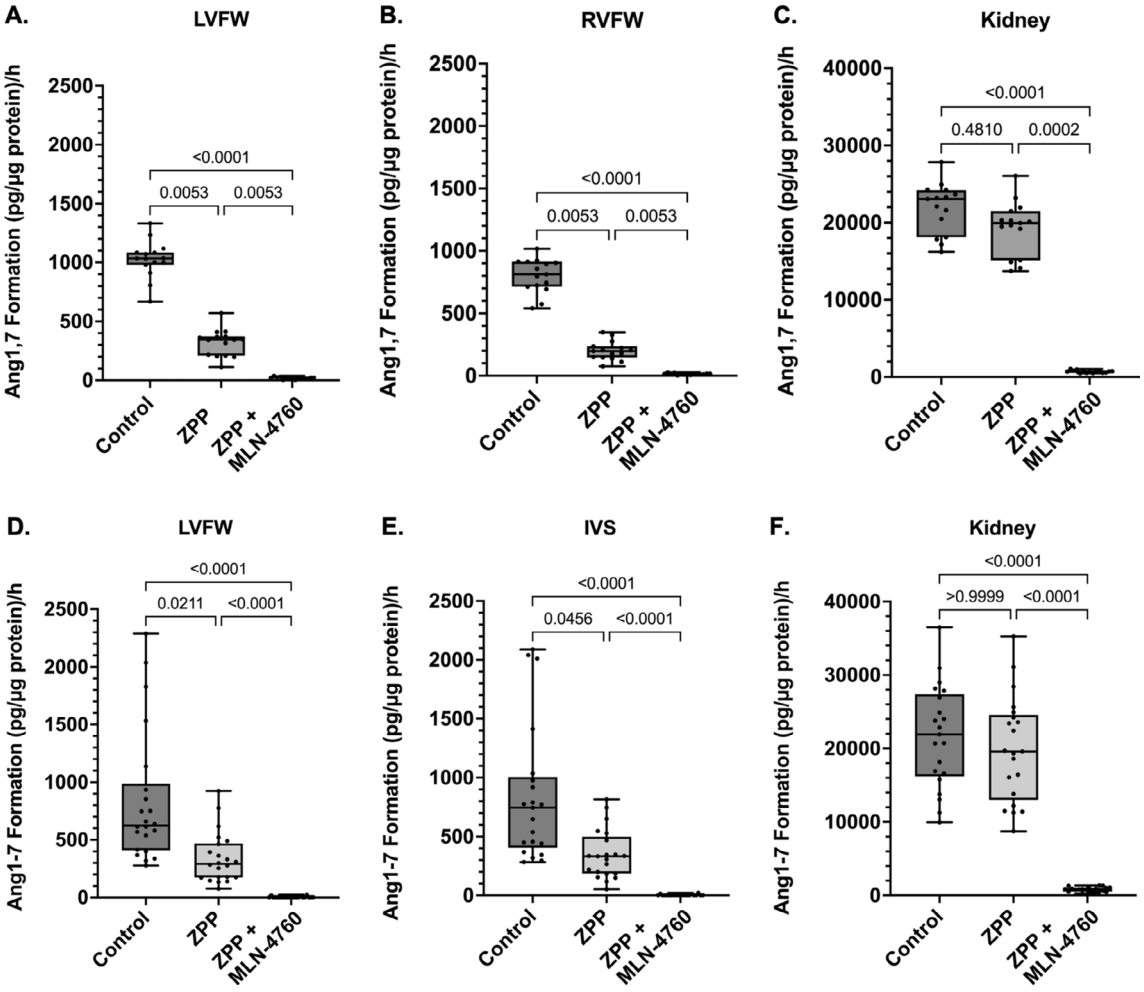

FIGURE 4 | Angiotensin 1–7 from angiotensin II formation rates ([pg/µg protein]/hour) in the myocardial and renal samples from dogs in Group 1 (A, B, and C) and Group 2 (D, E, and F). Homogenized tissues were incubated with angiotensin II alone (control) and with the prolyl-carboxypeptidase and prolylendopeptidase inhibitor Z-pro-prolinal (ZPP) or ZPP with the angiotensin-converting enzyme 2 inhibitor MLN-4760. Ang1-7, angiotensin 1–7; IVS, interventricular septum; LVFW, left ventricular free wall; RVFW, right ventricular free wall. *Note: p* values are adjusted for multiplicity; *p* values less than 0.01 are approximate.

included polycystic kidney disease and azotemia (one dog), azotemia and proteinuria (based on UP:C; two dogs), proteinuria alone (based on UP:C; one dog), and proteinuria alone without quantification (diagnosed by urine dipstick; two dogs). Three dogs were diagnosed postmortem with MMVD with mitral regurgitation (based on a historical left apical systolic murmur and myxomatous change to their mitral valve). Twelve dogs had a Whitney score of three, three dogs had a score of two, two dogs had a score of one, and five dogs had a score of zero. Cardiac medications included pimobendan (two dogs; Dogs 10 and 22), furosemide, and spironolactone (one dog; Dog 22). No dogs received any other RAS inhibitors. Four dogs (Dogs 4, 7, 10, and 19) were diagnosed postmortem with kidney disease based on histologic findings (Table S1).

4 | Discussion

RAS enzymes were present and active at 20 min postmortem in Group 1 and up to 40 h postmortem in Group 2. Relative enzyme activity could be estimated with sequential inhibition of enzymes and measurement of AP formation rates. These data serve as proof of concept for this method of assessing tissue RAS activity from samples collected and frozen minutes to hours rather than seconds postmortem.

Most, but not all, experiments had equivalent results when paired samples collected hours apart (Group 1 [0- and +2-h] and Group 2 [0- and +7-h]) were compared. Although many of the enzymes evaluated here were active for hours, the lack of a 100% equivalence between paired samples suggests that the timing of postmortem sampling affects enzyme activity estimates. Other factors that affect enzyme activity include temperature, degree of tissue degradation, and pH. Although the rate of equivalency between paired samples did not differ between the fresher samples of Group 1 and older samples of Group 2, consistent sample handling and uniform and rapid sample collection times will be necessary to reduce variation in future studies. Additionally, enzyme activities in the living organism change minute to minute in response to underlying physiological conditions, and these postmortem estimates more likely reflect the capacity or capability rather than the precise activity of the tissue RAS.

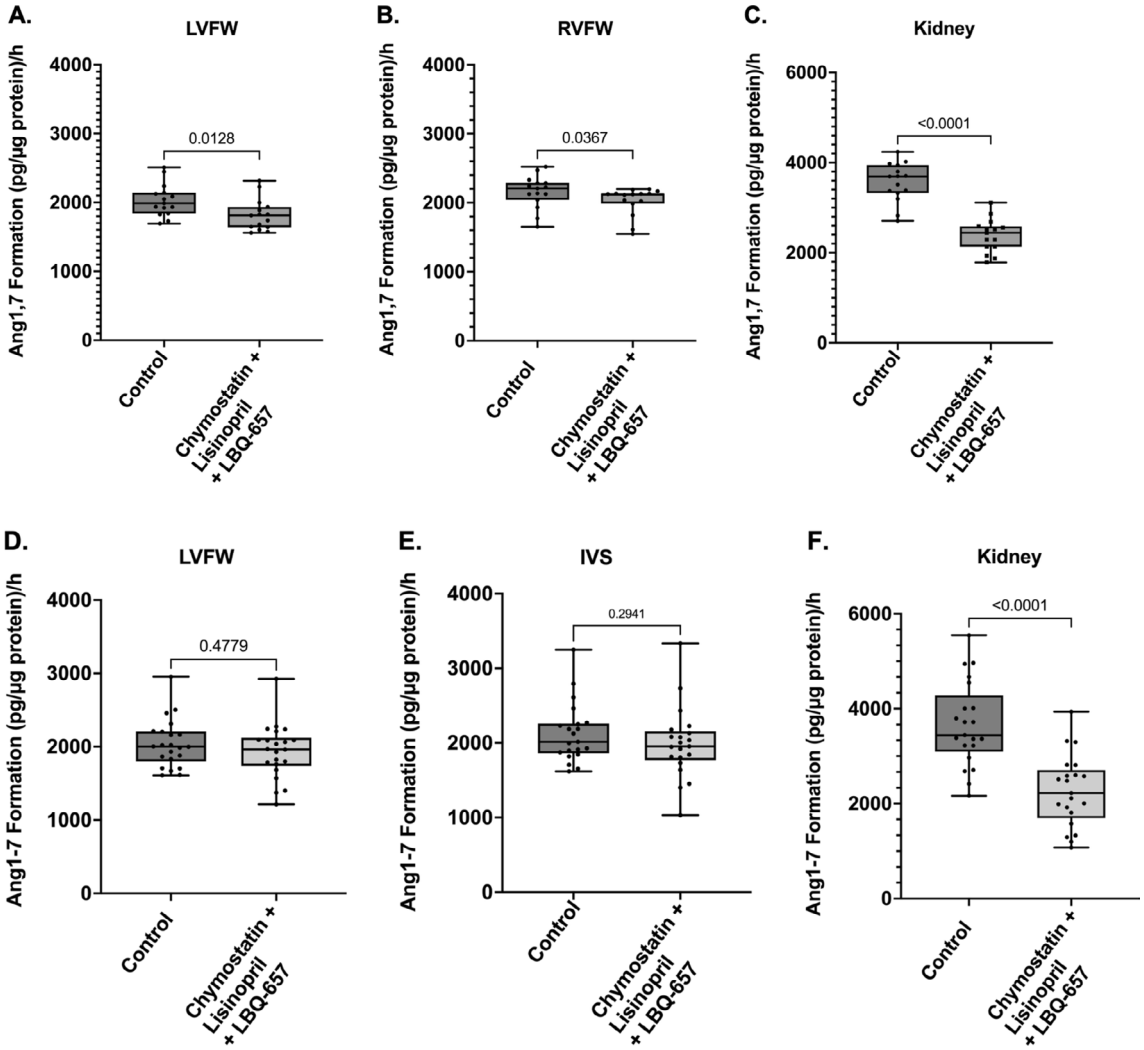

FIGURE 5 | Angiotensin 1–7 from angiotensin I formation rates ([pg/µg protein]/hour) in the myocardial and kidney samples from dogs in Group 1 (A, B, and C) and Group 2 (D, E, and F). Homogenized tissues were incubated with angiotensin I alone (control) and with chymostatin, lisinopril, and the neprilysin inhibitor LBQ-657. Ang1-7, angiotensin 1–7; IVS, interventricular septum; LVFW, left ventricular free wall; RVFW, right ventricular free wall. *Note: p* values are adjusted for multiplicity; *p* values less than 0.01 are approximate.

Myocardial AP concentrations were low postmortem and were not consistently measurable in quickly harvested (< 20 min) myocardial samples (Table 1). The LLOQ of AngI, AngII, and Ang1-7 for the LC–MS system used in this study was 10 fmol/g. This finding was similar to previous data of AP quantification in canine myocardium collected within 60 min of death (Oyama unpublished data). These findings in dogs differ from what has been found in pigs where myocardial AP concentrations were shown to be in the 15–20 fmol/g range and stable over the 1st hour postmortem [27]. The pigs used were both a control group and myocardial infarction model group. A prospective, comparative study would be required to determine if these differences in AP concentrations are due to species differences, time from death to sample freezing, or underlying disease.

Angiotensin peptide concentrations in the kidney were significantly higher than in the myocardium (Table 1). Directly measured postmortem AP concentrations in quickly extirpated renal tissues should be feasible to obtain in future studies. In the renal samples from Group 1, there was a moderate correlation between AngII concentrations and AngII production rates.

Endogenous enzyme activity in the postmortem samples was confirmed when all tissue homogenates incubated with AngI formed AngII and Ang1-7, and all tissue homogenates incubated with AngII formed Ang1-7. When interpreting the relative enzyme contributions to AngII and Ang1-7 formation, it must be acknowledged that RAS enzymes might vary in their stability postmortem, which could affect relative enzyme contribution. Also, the presence or absence of cardiovascular and renal disease varied among the dogs. That said, the enzyme activity results were similar between the more homogeneous Group 1 dogs and the heterogeneous Group 2 dogs, and some inferences regarding the postmortem RAS enzyme activity of these samples can be made.

Chymase inhibition did not significantly reduce AngII formation rates in any tissue, whereas the addition of the ACEi lisinopril to chymostatin inhibited AngII formation in all tissues (Figure 2). This indicates that ACE, not chymase, was responsible for the conversion of the supplied AngI substrate to AngII in these samples. Previous studies in dogs suggest that tissue enzyme dominance (chymase vs. ACE) varies depending on the

TABLE 4 | Contribution of various enzymes to angiotensin (Ang) II and Ang1-7 formation in samples of left ventricular free wall (LVFW), right ventricular free wall (RVFW), and kidney from 15 dogs (Group 1). Data are shown as median percent change (interquartile range).

Angiotensin production rate (pg/μg protein)/h	% Conversion by ACE	% Conversion by chymase	% Conversion by ACE + 2 h	% Conversion by chymase + 2 h
AngII formation from AngI: LVFW	88.6 (76.0–95.4)	11.4 (4.6–24.0)	93.3 (77.7–97.1)	6.7 (2.9–22.3)
AngII formation from AngI: RVFW	92.2 (85.0–97.1)	7.8 (2.9–15.0)	—	—
AngII formation from AngI: kidney	100 (97.5–100)	0 (0–2.5)	100 (96.9–100)	0 (0–3.1)

Angiotensin production rate (pg/μg protein)/h	% Conversion by PCP/PEP	% Conversion by ACE2	% Conversion other enzymes[a]	% Conversion by PCP/PEP + 2 h	% Conversion by ACE2 + 2 h	% Conversion other enzymes + 2 h[a]
Ang1-7 formation from AngII: LVFW	67.0 (63.0–77.7)	30.8 (20.9–35.0)	2.1 (1.4–2.6)	73.2 (65.3–79.9)	24.9 (19.8–33.1)	1.7 (1.2–2.0)
Ang1-7 formation from AngII: RVFW	76.8 (70.5–80.5)	21.5 (18.5–27.4)	1.7 (1.2–2.1)	—	—	—
Ang1-7 formation from AngII: kidney	12.0 (6.8–15.7)	84.7 (81.3–89.7)	3.3 (3.0–3.5)	8.7 (5.6–12.9)	88.9 (83.5–90.5)	3.2 (2.8–3.7)

Angiotensin production rate (pg/μg protein)/h	% Conversion by NEP	% Conversion by NEP + 2 h
Ang1-7 formation from AngI: LVFW	10.3 (6.7–14.8)	7.9 (4.3–9.5)
Ang1-7 formation from AngI: RVFW	5.8 (4.4–9.2)	—
Ang1-7 formation from AngI: kidney	32.03 (25.7–44.2)	36.3 (31.2–48.7)

Abbreviations: ACE, angiotensin-converting enzyme; ACE2, angiotensin-converting enzyme 2; NEP, neprilysin; PCP, prolyl-carboxypeptidase; PEP, prolyl-endopeptidase; ZPP, Z-pro-prolinal.
[a]The sum of PCP/PEP and ACE2% conversion does not reach 100%, as there was a small amount of Ang1-7 formation in the presence of both inhibitors; this Ang1-7 formation is accounted for as "other enzymes."

type and duration of hemodynamic overload [28–30]. Previous studies in people with end-stage heart disease and transplanted kidneys have demonstrated that chymase has a stronger affinity than ACE for AngI in renal and cardiac tissues, suggesting that when present, chymase is the primary mechanism for AngII formation in these tissues and disease states [24, 31, 32]. Angiotensin 1–12 (Ang1-12) has been purported to be an alternative precursor for AngII production via ACE and chymase. Determination of chymase- and ACE-mediated AngII formation from Ang1-12 would have required incubation with this precursor. However, Ang1-12 is not thought to occur naturally, does not appear to serve as an endogenous precursor for AngII [23], and was not accounted for in this study.

Angiotensin-converting enzyme 2 accounted for most of the AngII to Ang1-7 metabolism in the myocardial and renal samples, though PCP and PEP played a small role. Neprilysin also appeared important for AngI to Ang1-7 metabolism in the myocardium. Since neprilysin inhibition significantly reduced but did not completely prevent Ang1-7 formation in the renal tissues, other enzymes likely contributed to Ang1-7 formation in renal tissues. The myocardial findings differed from those in people with end-stage heart failure undergoing heart transplantation, where neprilysin and PCP, but not ACE2, were the primary Ang1-7-forming enzymes [32]. These differences might be due to dissimilarity in heart disease severity, sample age, and deterioration of enzyme activity, or species differences.

Generation of Ang1-7 from AngII in the renal samples was approximately 25 to 30 times higher than that in the myocardium. This finding aligns with an earlier study in dogs where directly measured ACE2 activity in the kidney was found to be between 250 and 500 times greater than myocardial tissue ACE2 activity [2]. In that study, the mean ACE2 activity in renal and myocardial tissues did not differ between dogs that were euthanized due to MMVD and dogs that died or were euthanized for non-cardiac reasons.

The ratio of Ang1-7 production (from AngII) and AngII production (from AngI) in control experiments (no inhibitors) revealed that the median Ang1-7 production was approximately 20-fold greater than AngII production in the myocardial tissues and approximately 30- to 50-fold greater than AngII production in the renal tissues (Tables 2 and 3). In all samples, Ang1-7 production was at least six-fold greater than AngII production, suggesting that the enzymes of the alternative RAS were dominant in these postmortem tissues.

TABLE 5 | Contribution of various enzymes to angiotensin (Ang) II and Ang1-7 formation in samples of left ventricular free wall (LVFW), interventricular septum (IVS), and kidney from 10 dogs (Group 2).

Angiotensin production rate (pg/μg protein)/h	% Conversion by ACE	% Conversion by chymase	% Conversion by ACE + 7 h	% Conversion by chymase + 7 h
AngII formation from AngI: LVFW	92.2 (86.3–95.5)	7.8 (4.5–13.7)	92.1 (86.8–96.1)	7.9 (4.0–13.2)
AngII formation from AngI: IVS	89.1 (81.7–94.3)	10.9 (5.7–18.3)	87.9 (81.4–97.7)	12.1 (2.3–18.7)
AngII formation from AngI: kidney	97.2 (96.0–100)	2.8 (0–4.0)	98.6 (85.5–100)	1.4 (0.1–14.5)

Angiotensin production rate (pg/μg protein)/h	% Conversion by PCP/PEP	% Conversion by ACE2	% Conversion other enzymes[a]	% Conversion by PCP/PEP + 7 h	% Conversion by ACE2 + 7 h	% Conversion other enzymes + 7 h[a]
Ang1-7 formation from AngII: LVFW	50.5 (23.7–85.0)	49.5 (15.0–76.3)	—	54.3 (21.0–80.6)	45.7 (19.4–79.0)	—
Ang1-7 formation from AngII: IVS	60.3 (25.4–81.6)	39.7 (18.4–74.6)	—	63.6 (21.7–79.0)	36.4 (21.0–78.3)	—
Ang1-7 formation from AngII: kidney	10.6 (4.2–15.4)	85.1 (80.9–92.7)	3.8 (2.8–4.4)	13.6 (6.1–16.1)	83.1 (80.5–89.9)	3.5 (2.8–4.3)

Angiotensin production rate (pg/μg protein)/h	% Conversion by NEP	% Conversion by NEP + 7 h
Ang1-7 formation from AngI: LVFW	4.2 (3.2–5.8)	5.9 (1.3–6.9)
Ang1-7 formation from AngI: IVS	4.2 (1.8–8.7)	5.8 (5.4–7.9)
Ang1-7 formation from AngI: kidney	37.2 (24.6–44.6)	35.3 (28.1–47.8)

Note: Data shown as median percent change (interquartile range); one dog did not have a kidney sample collected.
Abbreviations: ACE, angiotensin-converting enzyme; ACE2, angiotensin-converting enzyme 2; NEP, neprilysin; PCP, prolyl-carboxypeptidase; PEP, prolyl-endopeptidase; ZPP, Z-pro-prolinal.
[a]The sum of PCP/PEP and ACE2% conversion does not reach 100%, as there was a small amount of Ang1-7 formation in the presence of both inhibitors; this Ang1-7 formation is accounted for as "other enzymes."

TABLE 6 | Angiotensin (Ang) production rate for the 2-h samples under various conditions for left ventricular free wall (LVFW) and kidney from 15 dogs (Group 1); median (interquartile range). For the baseline sample results, see Table 2.

Angiotensin production rate (pg/μg protein)/h	Control + 2 h	Chymostatin + 2 h	Chymostatin + lisinopril + 2 h
AngII formation from AngI: LVFW	43 (30–48)	43 (27–45)	All < 5
AngII formation from AngI: kidney	640 (597–852)	661 (499–875)	All < 5

Angiotensin production rate (pg/μg protein)/h	Control + 2 h	ZPP + 2 h	ZPP + MLN-4760 + 2 h
Ang1-7 formation from AngII: LVFW	956 (875–1073)	226 (194–334)	15 (11–22)
Ang1-7 formation from AngII: kidney	22 548 (19949–22 910)	19 728 (18869–21 373)	701 (553–783)

Angiotensin production rate (pg/μg protein)/h	Control + 2 h	Chymostatin + lisinopril + LBQ-657 + 2 h
Ang1-7 formation from AngI: LVFW	1867 (1797–2015)	1741 (1626–1859)
Ang1-7 formation from AngI: kidney	3476 (3280–3784)	1931 (1809–2397)

Note: Angiotensin II and 1–7 formation rates did not differ significantly between the immediately frozen samples and the samples where freezing was delayed for 2 h.
Abbreviation: ZPP, Z-pro-prolinal.

TABLE 7 | Angiotensin (Ang) production rate under various conditions for left ventricular posterior wall (LVPW), interventricular septum (IVS), and kidney from 10 of the 22 dogs in Group 2 that had a second sample taken 7 h later (+7 h); median (interquartile range). One dog did not have a kidney sample collected.

Angiotensin production rate (pg/μg protein)/h	Control	Chymostatin	Chymostatin + lisinopril	Control + 7 h	Chymostatin + 7 h	Chymostatin + lisinopril + 7 h
AngII from AngI: LVPW	26 (24–31)	25 (21–28)	All dogs < 5	29 (27–33)[a]	27 (25–31)[a]	All dogs < 5
AngII from AngI: IVS	30 (34–38)	25 (21–35)	All dogs < 5	30 (27–36)	27 (22–34)	All dogs < 5
AngII from AngI: kidney	366 (190–1082)	339 (185–1137)	All dogs < 5	409 (195–1013)	341 (194–850)	All dogs < 5
Angiotensin production rate (pg/μg protein)/h	**Control**	**ZPP**	**ZPP + MLN-4760**	**Control + 7 h**	**ZPP + 7 h**	**ZPP + MLN-4760 + 7 h**
Ang1-7 from AngII: LVPW	597 (333–874)	178 (145–301)	All but 1 dog < 5 (18.9)[b]	619 (379–777)	224 (172–308)	All but 1 dog < 5 (15.0)[b]
Ang1-7 from AngII: IVS	592 (424–822)	247 (141–346)	All but 1 dog < 5 (15.4)[b]	609 (483–766)	236 (161–372)	All but 1 dog < 5 (17.5)[b]
Ang1-7 from AngII: kidney	22 860 (14852–28 014)	19 707 (12664–24 561)	917 (504–1150)	21 281 (15246–27 783)	19 174 (12681–25 532)	890 (433–1183)

Angiotensin production rate (pg/μg protein)/h	Control	Chymostatin + lisinopril + LBQ-657	Control + 7 h	Chymostatin + lisinopril + LBQ-657 + 7 h
Ang1-7 from AngI: LVPW	2013 (1876–2177)	1945 (1796–2086)	2141 (1975–2207)	2049 (1963–2147)
Ang1-7 from AngI: IVS	2156 (1902–2256)	2021 (1892–2102)	2176 (1808–2291)	1982 (1709–2133)
Ang1-7 from AngI: kidney	3795 (3293–4476)	2514 (1965–2812)	3969 (3397–4544)	2399 (1882–2753)

Abbreviation: ZPP, Z-pro-prolinal.
[a]Differed significantly when compared to the initial sample (Wilcoxon matched-pairs signed rank test).
[b]Values for individual dogs with angiotensin concentrations greater than the lower limit of quantification.

TABLE 8 | Comparison between angiotensin (Ang) production rate for the 0- and 2-h samples under various conditions for left ventricular free wall (LVFW), right ventricular free wall (RVFW), and kidney from 15 dogs (Group 1).

Angiotensin production rate (pg/μg protein)/h	Control: comparison 0 h and +2 h geometric mean	90% CL lower bound > 0.8 upper bound < 1.2	Equivalence assessment	Chymostatin: comparison 0 h and +2 h: geometric mean	90% CL lower bound > 0.8 upper bound < 1.2	Equivalence assessment
AngII formation from AngI: LVFW	1.0302	0.9816–1.0926	Equivalent	1.0117	0.9580–1.0683	Equivalent
AngII formation from AngI: kidney	0.8956	0.7955–1.000	Not equivalent	0.9037	0.8018–1.0186	Equivalent

Angiotensin production rate (pg/μg protein)/h	Control: comparison 0 h and +2 h geometric mean	90% CL Lower bound > 0.8 Upper bound < 1.2	Equivalence assessment	ZPP: comparison 0 h and +2 h: geometric mean	90% CL Lower bound > 0.8 Upper bound < 1.2	Equivalence assessment	ZPP + MLN-4760: comparison 0 h and +2 h: geometric mean	90% CL Lower bound > 0.8 Upper bound < 1.2	Equivalence assessment
Ang1-7 formation from AngII: LVFW	1.0565	1.0119–1.1031	Equivalent	1.1958	1.1102–1.2881	Not Equivalent	1.1361	0.9555–1.3509	Not equivalent
Ang1-7 formation from AngII: kidney	1.0020	0.9699–1.0351	Equivalent	0.9658	0.9231–1.0104	Equivalent	1.0241	0.9295–1.1283	Equivalent

Angiotensin production rate (pg/μg protein)/h	Control: comparison 0 h and +2 h geometric mean	90% CL Lower bound > 0.8 Upper bound < 1.2	Equivalence assessment	Chymostatin + lisinopril + LBQ-657: Comparison 0 h and +2 h: geometric mean	90% CL Lower bound > 0.8 Upper bound < 1.2	Equivalence assessment
Ang1-7 formation from AngI: LVFW	1.0559	1.0148–1.0987	Equivalent	1.0213	0.9924–1.0510	Equivalent
Ang1-7 formation from AngI: kidney	1.0285	1.0069–1.0506	Equivalent	1.1259	1.0388–1.2202	Not equivalent

Note: For the Ang production rate results, see Tables 2 and 4 and Figures 3–5. The geometric mean of the ratio of the 0- and 2-h samples is reported along with the 90% confidence limit (CL) for the lower and upper bounds. The criteria for equivalence are a lower bound > 0.8 and an upper bound < 1.2. Angiotensin II and 1–7 formation rates did not differ significantly between the immediately frozen samples and the samples where freezing was delayed for 2 h.
Abbreviation: ZPP, Z-pro-prolinal.

TABLE 9 | Comparison between angiotensin (Ang) production rate for the 0- and 7-h samples under various conditions for left ventricular free wall (LVFW), interventricular septum (IVS), and kidney from 10 dogs (Group 2).

Angiotensin production rate (pg/μg protein)/h	Control: comparison 0 h and +7 h geometric mean	90% CL lower bound > 0.8 upper bound < 1.2	Equivalence assessment	Chymostatin: comparison 0 h and +7 h: geometric mean	90% CL lower bound > 0.8 upper bound < 1.2	Equivalence assessment
AngII formation from AngI: LVFW	0.8525	0.7647–0.9504	Not equivalent	0.8580	0.7689–0.9574	Not equivalent
AngII formation from AngI: IVS	0.9588	0.8910–1.0318	Equivalent	0.9456	0.8695–1.0284	Equivalent
AngII formation from AngI: kidney	0.9730	0.8611–1.0994	Equivalent	1.0570	0.9075–1.2312	Not equivalent

Angiotensin production rate (pg/μg protein)/h	Control: comparison 0 h and +7 h geometric mean	90% CL Lower bound > 0.8 Upper bound < 1.2	Equivalence assessment	ZPP: Comparison 0 h and +7 h: geometric mean	90% CL Lower bound > 0.8 Upper bound < 1.2	Equivalence assessment	ZPP + MLN-4760: comparison 0 h and +7 h: geometric mean	90% CL Lower bound > 0.8 Upper bound < 1.2	Equivalence assessment
Ang1-7 formation from AngII: LVFW	0.9714	0.8985–1.0502	Equivalent	0.9494	0.8683–1.0381	Equivalent	—	—	—
Ang1-7 formation from AngII: IVS	0.9427	0.8284–1.0729	Equivalent	0.9257	0.8255–1.0380	Equivalent	—	—	—
Ang1-7 formation from AngII: kidney	1.0220	0.9308–1.1222	Equivalent	1.0407	0.9289–1.1659	Equivalent	1.0421	0.9357–1.1605	Equivalent

Angiotensin production rate (pg/μg protein)/h	Control: comparison 0 h and +7 h geometric mean	90% CL Lower bound > 0.8 Upper bound < 1.2	Equivalence assessment	Chymostatin + Lisinopril + LBQ-657: comparison 0 h and +7 h: geometric mean	90% CL Lower bound > 0.8 Upper bound < 1.2	Equivalence assessment
Ang1-7 formation from AngI: LVFW	0.9661	0.9285–1.0051	Equivalent	0.9678	0.9249–1.0127	Equivalent

(Continues)

TABLE 9 | (Continued)

Angiotensin production rate (pg/μg protein)/h	Control: comparison 0 h and +7 h geometric mean	90% CL Lower bound > 0.8 Upper bound < 1.2	Equivalence assessment	Chymostatin + Lisinopril + LBQ-657: comparison 0 h and +7 h: geometric mean	90% CL Lower bound > 0.8 Upper bound < 1.2	Equivalence assessment
Ang1-7 formation from AngI: IVS	1.0133	0.9707–1.0579	Equivalent	1.0348	0.9785–1.0944	Equivalent
Ang1-7 formation from AngI: kidney	1.0002	0.9695–1.0320	Equivalent	1.0259	0.9935–1.0594	Equivalent

Note: One dog did not have a kidney sample collected. For the Ang production rate results, see Tables 3 and 5 and Figures 3–5. The geometric mean of the ratio of the 0- and 7-h sample is reported along with the 90% confidence limit (CL) for the lower and upper bounds. The criteria for equivalence are a lower bound > 0.8 and upper bound < 1.2. Angiotensin II and 1–7 formation rates did not differ significantly between the immediately frozen samples and the samples where freezing was delayed for 7 h.
Abbreviation: ZPP, Z-pro-prolinal.

This study has strengths and limitations. This study evaluated the stability of comprehensive RAS enzyme activity in canine myocardial and renal tissues. Although the sample size is small, it is larger than what would have been possible using endomyocardial or renal biopsy samples obtained during the same time period. The ability to compare myocardial and renal tissues from the same animal and to compare two myocardial sites is also valuable and would be difficult to do with biopsy samples. A key limitation in the Group 2 samples is the inability to assess the effect of time on RAS enzyme activity due to the variation in the timing of postmortem sample collection among dogs. Variation in the amount of microvasculature and circulating RAS enzymes and peptides contained within samples was also likely present. Additionally, some dogs died naturally, while others were humanely euthanized with an overdose of pentobarbital, which might have affected local RAS activity. Finally, dogs in Group 2 were enrolled as they presented to our pathology service and were a heterogeneous cohort with variation in age, sex, neuter status, underlying disease process, and historical medical therapies. The small sample size and heterogeneity of diseases in the dogs of Group 2 precluded a robust statistical analysis, and firm conclusions about the effect of disease on the tissue RAS could not be made.

RAS enzymes were present and active for up to 40 h postmortem and relative enzyme contribution could be estimated. Angiotensin peptide concentrations in postmortem samples were low or below the LLOQ, even in samples extirpated within 20 min of euthanasia. The activity of Ang1-7-forming enzymes dominated over AngII-forming enzymes in these postmortem samples. The renal tissues demonstrated a greater capacity for AngI to AngII and AngII to Ang 1–7 turnover as compared to the myocardial tissues. Given the limitations inherent to using postmortem tissues, these conclusions cannot be applied to the antemortem AP concentrations and tissue enzyme activity. Understanding the behavior of the tissue RAS during cardiac and renal disease and during pharmacologic RAS modulation will improve our ability to identify future therapeutic strategies to treat cardiac and renal disease in dogs and cats. The identification of differences between these species and humans might also generate further hypotheses for translational studies.

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Ethics Statement

Dogs in Group 1 were under the care of TRS Labs Inc. (Athens, Georgia), which adheres to the principles outlined in the Guide for the Care and Use of Laboratory Animals. The Institutional Animal Care and Use Committee (IACUC) approval number was 22-06. Dogs in Group 2 did not require IACUC approval, as owners gave the University of California, Davis Veterinary Medical Teaching Hospital permission

for 'unrestricted use' during the necropsy examination. The authors declare that human ethics approval was not needed.

Conflicts of Interest

Ames has received consulting fees, research support, and honoraria from Ceva Sante Animale and honoraria from Boehringer Ingelheim. Domenig is an employee of Attoquant Diagnostics, a fee-for-service laboratory. Weitzhandler and Wilson have no relevant conflicts of interest to declare.

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